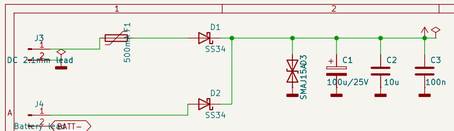
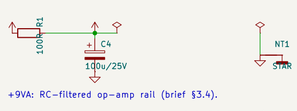
POWER — parallel diode-OR (pinmap §4n), no regulator. VIN abs max 17 V: 12 V adapters yes, 18 V no.



Rail = VIN (≈8.6 V battery / ≈11.6 V adapter). Battery − returns via the OUT ring (the power switch).



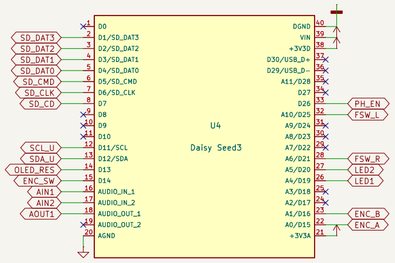
+9VA: RC-filtered op-amp rail (brief §3.4).



GROUND STAR — NT1 is the single AGND↔DGND join (pinmap §4l). Place at the power-entry corner.

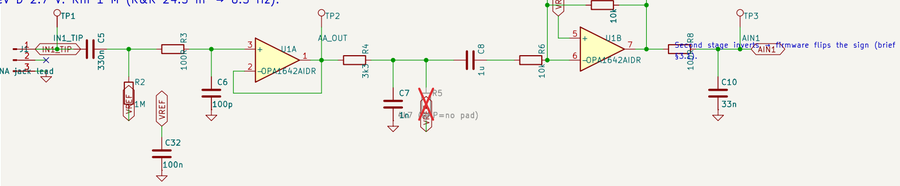


DAISY SEED3 — pin numbers per daisy/board.h. Unused GPIO marked NC.

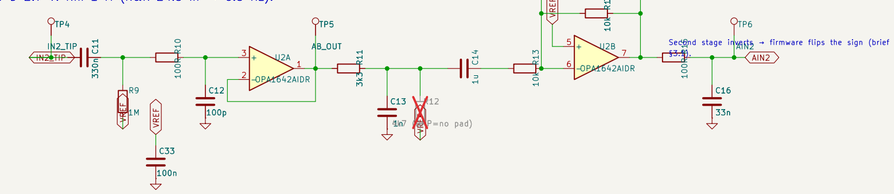


USB: the Seed's own USB-C port via the panel extender — no board connection (D29/D30 NC).

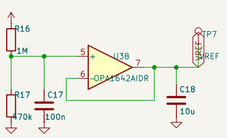
FRONT END CH A — Seed3 Fig 3.3 (RF stopper · unity follower · pole+pad · inverting unity · 100R+33n), Vref = rev D 2.7 V. Rin 1 M (K&K 24.5 nF → 6.5 Hz).



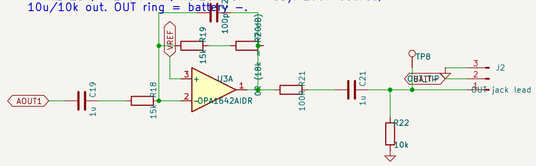
FRONT END CH B — Seed3 Fig 3.3 (RF stopper · unity follower · pole+pad · inverting unity · 100R+33n), Vref = rev D 2.7 V. Rin 1 M (K&K 24.5 nF → 6.5 Hz).



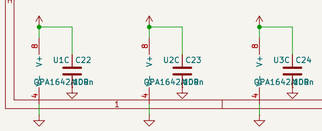
XLR MIC FRONT-END + 48 V PHANTOM — CH2 combo: XLR mic (INA163, G=1+6k/RG, RG=120R=+34 dB fixed + DSP) OR 1/4" stomp-box/instrument via the combo's normalling switch. LM2577 boost → 48 V phantom, GPIO P-FET enable (default OFF). Single-supply off VIN (best at 12 V).
VREF — rev D bias: 1 M / 470 k from +9VA → 2.7 V. Buffered by U3B.



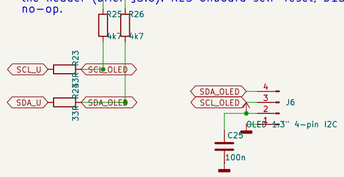
OUTPUT — Seed3 Fig 3.6 instrument-level stage, UNITY (Rin = Rf1 15k, Rf2 OR link; fit 18k for +7 dB). 100R source, 10u/10k out. OUT ring = battery −.



OP-AMP POWER — +9VA / AGND, 100n at each chip.

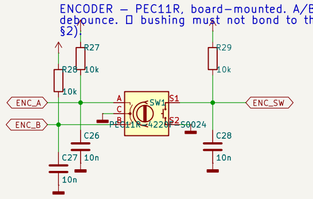


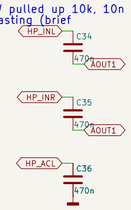
OLED 1.3" SH1106 I²C module, 4-pin (GND/VCC/SCL/SDA) → 0x3C (board.h). 33R series at the Seed, 4k7 pull-ups at the header (brief §3.6). RES onboard self-reset, D13 a no-op.

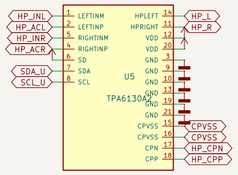


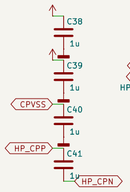
HEADPHONE MONITOR — TPA6130A2 I2C DirectPath (0x60, no output caps). Taps AOUT1 dual-mono to a 3.5mm panel jack. VDD=+3V3D (charge-pump off +3V3A). Volume/mute over I2C = firmware menu.

ENCODER — PEC11R, board-mounted. A/B/SW pulled up 10k, 10n debounce. ⌀ bushing must not bond to the casting (brief §2).



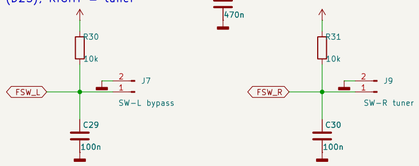








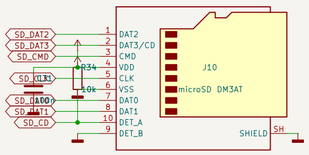
FOOTSWITCHES — panel-mounted momentary SPST on JST 2-pin leads. 10k pull-up + 100n. LEFT = bypass (D25), RIGHT = tuner (D21).



STATUS LEDs — D19 = bypass/engaged, D20 = looper/record. 3 mm through the face; shape/position distinguish them, not colour.

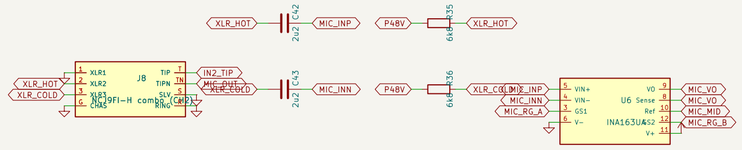


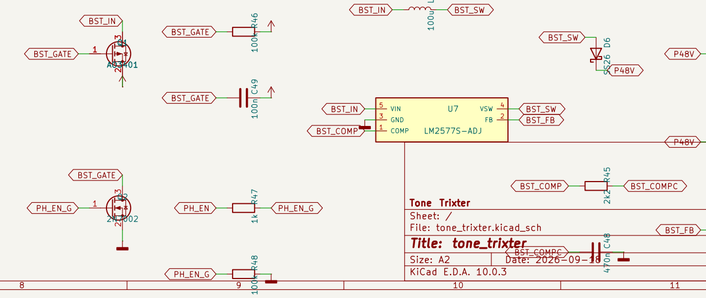
microSD — push-push socket ON THE SDMMC PINS (D1-D6). Bit-bang SPI fallback = 4 pin numbers in sd_spi.h: CS=D1 MOSI=D5 SCK=D6 MISO=D4 (brief §3.7).



TEST POINTS — every pad has an adjacent GND pad on the layout (feedback_design_for_probeability).







Tone Trixter
Sheet: /
File: tone_trixter.kicad_sch
Title: tone_trixter
Size: A2 | Date: 2026-09-30 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 1/1